

A Three-Family Diagnostic Audit for Train-Free Prior-Override Verification in Vision-Language Models

Anonymous submission

Abstract

Vision-language models can answer from learned language and scene regularities even when an image contains evidence that should override those priors. Existing prior-conflict and modality-conflict work motivates the problem, but a diagnostic paper also needs prompt-format controls, image-blind controls, and a ledger that prevents scoped comparisons from becoming broader claims. We report a train-free audit of Qwen2.5-VL-3B-Instruct, a named frozen open-weight VLM, on three executable image-backed families: TextVQA/Open Images, VQA v2/COCO, and RefSpatial/Location. The primary run contains 1050 scored model-output rows: 900 B0–B5 matrix rows and 150 B6 log-probability comparator/control rows; a separate source-clean guardrail file contains 300 proxy-control scored rows. TextVQA and VQA v2 use exact literal-answer accuracy, while RefSpatial uses source-mask point-hit scoring. Image-blind B4 is 0.100000 on TextVQA, 0.220000 on VQA v2, and 0.000000 on RefSpatial, showing that question priors still solve some rows but not all families equally. Evidence-grounded B5 is mixed rather than a robust mitigation: it improves TextVQA over the matched-format B1 baseline by 0.080000, is net zero on VQA v2, and is -0.040000 on RefSpatial. B6 is an executed scoped control with scores 0.820000, 0.900000, and 0.180000 on the same three families, so the contribution is a calibrated diagnostic protocol and evidence ledger rather than a mechanistic superiority claim.

Introduction

Vision-language models (VLMs) are often evaluated as if the image is the decisive evidence, yet recent diagnostics show that answers can follow language priors, textual shortcuts, and scene expectations when modalities disagree. ViLP constructs visual-language conflicts and observes that VLMs may favor learned priors over visible evidence (Luo et al. 2025). Modality-preference work shows that textual variation can dominate visual evidence in vision-centered tasks (Deng et al. 2025). Bias and counterfactual probes similarly study answer shifts when visual and language evidence are placed in tension (Vo et al. 2025). These findings make prior override an evaluation problem before it is a mitigation problem.

The difficulty is that prompt changes are easy to overinterpret. A prompt that asks a model to inspect visible evidence can improve a row because it changes the answer format,

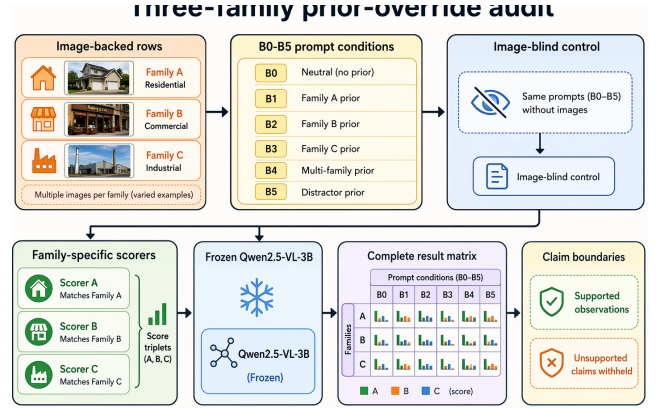


Figure 1: Diagnostic scope. The audit separates prior-biased answers, image-grounded checks, and scoped comparator evidence; numerical claims come from the result tables rather than from this conceptual overview.

because it induces different reasoning verbosity, or because the row was answerable from the question alone (Wu et al. 2025; Li et al. 2025b). Conversely, a direct image-present score does not prove grounding unless an image-blind condition is measured (Luo et al. 2025; Deng et al. 2025; Vo et al. 2025). A diagnostic for train-free prior override therefore needs direct answering, matched-format answering, image-blind answering, and an evidence-grounded variant in the same row ledger. It also needs explicit labels for missing or blocked comparisons, because a missing benchmark family or comparator is a scientific limitation rather than a prose problem.

This paper instantiates that diagnostic with a new three-family audit of Qwen2.5-VL-3B-Instruct. The evaluated model is named, frozen, and open-weight; no training, adapter, retrieval index, reinforcement learning, or checkpoint update is used. The retained families are TextVQA/Open Images, VQA v2/COCO, and RefSpatial/Location. TextVQA and VQA v2 are scored by exact literal-answer matching against source annotations. RefSpatial is scored by whether the generated point falls inside the source segmentation mask for the referred target. Each family contributes 50 executable image-backed tasks, ev-

ery task is evaluated under B0–B5, and B6 is replayed as a scoped log-probability comparator/control over the same 150 retained rows.

The measured pattern is useful because it is not uniformly positive. TextVQA is favorable to the evidence-grounded prompt relative to the matched-format baseline: B5 reaches 0.800000 versus B1 at 0.720000. VQA v2 is already strong under matched-format prompting: B1 and B5 both score 0.880000, with two recoveries and two regressions. RefSpatial is weak for this model and prompt family: B1 scores 0.260000 and B5 scores 0.220000. B6 matches or exceeds B5 on TextVQA and VQA v2 but is weaker on RefSpatial. B4 is also informative. It scores 0.100000 on TextVQA, 0.220000 on VQA v2, and 0.000000 on RefSpatial, which means some retained rows remain answerable from question priors alone.

This mixed pattern is the reason the paper is framed as an audit. The goal is not to maximize one prompt recipe on a small slice; it is to make the reader see which conclusions survive when the same rows are run with image-present, image-blind, matched-format, and evidence-grounded variants (Zhao et al. 2025; Ren et al. 2026; Guo et al. 2026). A single score would hide the scientific issue. Text recognition, everyday object questions, and spatial pointing all involve image evidence, but they expose different shortcuts and different scorer assumptions (Li et al. 2025c; Liu et al. 2025). The paper therefore asks a narrower question: when a named frozen VLM is replayed under a fixed B0–B5 protocol, which claims about prior override are visible, which are contradicted, and which remain outside the evidence?

The contribution is a calibrated audit rather than a broad mitigation claim. First, the paper provides a single family-by-condition matrix covering every retained family, B0–B5 method cell, and scoped B6 comparator/control for a publishably identifiable VLM. Second, it shows that image-blind controls are not merely formal: they vary sharply across TextVQA, VQA v2, and RefSpatial. Third, it uses a claim ledger to keep broad B5-improvement, transformed no-conflict safety, GQA headline, and mechanistic-superiority claims out of the paper when the evidence does not support them. Figure 1 summarizes the scope.

Related Work

Visual-language-prior diagnostics are the closest setting for this audit. ViLP studies deliberate prior conflicts in VLMs (Luo et al. 2025). Modality-conflict work analyzes cases where VLMs place excessive trust in text over visual evidence (Deng et al. 2025). Counterfactual bias probes ask how VLM behavior changes under controlled shifts in visual or language evidence (Vo et al. 2025). Language-prior studies further show that shortcut correlations can shape behavior even when images are available (Wu et al. 2025; Li et al. 2025b). The present paper uses these studies to motivate controls that separate ordinary VQA accuracy from prior-sensitive behavior.

Those studies also motivate a conservative standard for evidence. A conflict benchmark can show that a model sometimes follows the wrong modality, but it does not by itself show that a mitigation prompt caused the repair (Luo

et al. 2025; Vo et al. 2025). Prompting changes can alter formatting, verbosity, refusal behavior, and answer priors at the same time (Wu et al. 2025; Deng et al. 2025). For that reason, this audit treats a direct answer, a matched-format answer, and an evidence-grounded answer as different observations rather than as a clean causal ladder. The B1–B5 comparison is the closest available contrast because both rows require the same compact answer field, but even that comparison remains a prompt-level diagnostic rather than a mechanism proof.

Hallucination and mitigation work motivates the train-free verification framing. Language-prior studies argue that VLMs can rely on shortcut correlations in image-conditioned tasks (Wu et al. 2025; Li et al. 2025b). Contrastive and decode-time methods try to reduce hallucinated or prior-dominated outputs by comparing contexts or representations (Zhao et al. 2025; Ren et al. 2026; Guo et al. 2026). Causal and attention-oriented work suggests that shortcut behavior may persist even when relevant visual evidence is represented (Li et al. 2025a; Liu et al. 2025). Our audit is narrower: it does not train a mitigation, and its B6 log-probability control is reported descriptively rather than as a mechanism-level contrastive repair. The audit checks whether conservative evidence-grounded and comparator conditions change answers under a fixed frozen model.

This distinction matters for how results should be read. A train-free prompt can be useful as an evaluation probe even when it is not a reliable deployment intervention (Zhao et al. 2025; Ren et al. 2026). If B5 recovers a row that B1 misses, the row becomes a candidate example where explicit visual-evidence language changed the answer. If B5 regresses, the same intervention may have disrupted an otherwise correct answer. If B1 and B5 tie, the result may indicate either stable evidence use or an insensitive prompt contrast. The paired rows in this paper contain all three possibilities, so the related mitigation literature is background for the design rather than evidence that B5 should win.

Context-prior work is also relevant because scene priors are not inherently wrong. Human and machine perception can benefit from reliable context (Torralba 2003; Bar 2004; Oliva and Torralba 2007). The diagnostic question is whether the model can privilege the literal image when a plausible prior should be overridden. Causal and attention-oriented studies raise the related possibility that a model may encode relevant evidence but still choose the final answer through a shortcut (Li et al. 2025b,a; Liu et al. 2025). Future versions should therefore add region, crop, or counterfactual checks, not only answer strings (Li et al. 2025c).

The three retained families instantiate this positioning at different levels of granularity. TextVQA is not a pure prior-conflict benchmark, but visible strings make question-only success difficult on the retained rows. VQA v2 is a standard visual question answering source with stronger dataset regularities, making it a useful stress test for image-blind controls. RefSpatial is a spatial-reference source whose point metric is harsher and not directly comparable to exact-answer VQA (Li et al. 2025c; Liu et al. 2025). The paper deliberately keeps these differences visible instead of forcing them into a single combined score.

Diagnostic Benchmark and Setup

The audit uses three retained families. TextVQA/Open Images emphasizes visible text and short strings. VQA v2/COCO emphasizes object, count, color, activity, and scene questions with answer annotations. RefSpatial/Location tests spatial reference by asking for a point on a referred object or region, then scoring the point against a source mask. RefSpatial replaces GQA/Visual Genome in this evidence pass because the available GQA slice did not provide source images in an evaluable form for the same replay protocol. Including those GQA rows would mix benchmark construction failure with model behavior.

The retained-family decision is part of the method. A family is included only when the evaluation can bind the prompt, image, and scorer target in a replayable way. For TextVQA and VQA v2, that means a source image and answer set can be paired with a short model response. For RefSpatial, it means the referred target has a mask that can score a generated point. The excluded GQA branch is not treated as negative model evidence, because the failure is upstream of model evaluation: the rows available for this pass did not satisfy the same image-backed replay contract. That exclusion is reported as an evidence gap rather than hidden inside the benchmark name, following the same claim-calibration logic used in prior-conflict and counterfactual diagnostics (Luo et al. 2025; Vo et al. 2025; Li et al. 2025c).

The evaluated backend is Qwen2.5-VL-3B-Instruct. The model is held fixed for all families and prompt conditions. Decoding is greedy with a 64-token cap. The audit is inference-only: no weights, adapters, prompt-tuned parameters, retrieval index, verifier model, or policy are trained or updated. This makes the result interpretable as a diagnostic of one named frozen VLM under one replayable protocol, not as a claim about trained adaptation.

Each source row is evaluated under six conditions. B0 direct answering asks for the answer directly. B1 matched format uses the same image but requires a compact JSON answer, controlling for answer schema. B2 self-verification asks the model to verify before returning the final field. B3 reasoning with final-field extraction permits brief reasoning but scores only the final answer field. B4 image-blind text-only control removes the image and keeps the task prompt. B5 evidence-grounded answering keeps the image and asks the model to prioritize visible evidence before returning the same compact answer form.

The metrics are deterministic and source-grounded. TextVQA and VQA v2 outputs are parsed as JSON or first-line answers, normalized, and matched against the normalized source answer list. RefSpatial outputs are parsed as a point; points expressed on a 0–999 scale are mapped to image coordinates and counted correct when they land inside the source mask. The RefSpatial metric is not the same as VQA exact answer accuracy, so the paper reports metrics by family rather than pooling them into one headline average (Li et al. 2025c; Liu et al. 2025).

Parsing is intentionally simple. For VQA-style rows, the scorer looks for the compact answer field when present and otherwise falls back to the first concise answer span. This allows B0, B1, B2, B3, and B5 to be scored under the same

answer-normalization rule while preserving the prompt-role differences. For RefSpatial, the scorer accepts numeric point outputs and maps them into the target image coordinate system before the mask check. Non-point or unparsable RefSpatial answers are failures. These choices favor transparency over semantic adjudication: they reduce evaluator discretion, but they also mean that paraphrases, near misses, and partially correct spatial descriptions are not credited.

The retained rows are selected for replayability. A row enters the audit only if the source question is nonempty, the image is available for evaluation, and the scorer has source-derived ground truth. This produces 50 tasks per family, the current executable maximum recorded for the retained source slices. The slice is still modest and should not be read as distributionally representative. That limitation is reflected in the claim ledger: the evidence can support a diagnostic comparison, but it cannot support broad source-general or robust mitigation claims.

The six prompt conditions are designed to separate three confounds. The first is answer format: B1 and B5 share compact structured output, while B0 does not. The second is reasoning verbosity: B2 and B3 allow verification or reasoning language but are still scored through a final answer field. The third is image dependence: B4 removes the image while keeping the question. This design cannot prove that the model attends to a particular region, but it can prevent a common error in interpretation. If an image-present score is high and the image-blind score is also high, the row cannot support a strong claim that visual evidence was necessary for that answer (Deng et al. 2025; Wu et al. 2025).

Train-Free Verification Method

Figure 2 shows the audit pipeline. The input is a source row containing an image, a task prompt, and source-derived ground truth. The row is expanded into the B0–B5 conditions and evaluated once per condition. The scorer records the raw output, parsed prediction, score, family, method, and metric. Aggregation is performed by family and method, not across heterogeneous metrics.

B1 and B5 are the main prompt-effect comparison. B0 and B5 differ in both answer role and answer schema, so a B0–B5 contrast can confound direct-answer behavior with evidence-grounding instructions. B1 and B5 share the compact answer format, which makes their paired comparison the cleanest available slice for the evidence-grounding instruction. Even this comparison is not a causal decomposition: prompt wording can alter multiple latent behaviors (Wu et al. 2025; Zhao et al. 2025; Ren et al. 2026). The paper therefore treats B1–B5 as a controlled diagnostic comparison, not as proof of a mechanism.

B4 is the image-dependence check. If B4 succeeds, the row may be answerable from language priors or dataset regularities alone. If B4 fails while image-present variants succeed, the row better supports an image-dependence statement (Deng et al. 2025; Wu et al. 2025; Vo et al. 2025). The new evidence shows both cases. TextVQA has B4 at 0.100000, VQA v2 has B4 at 0.220000, and RefSpatial has B4 at zero. Those blind-control successes prevent the paper

Method Pipeline: Train-Free Prior-Override Verification

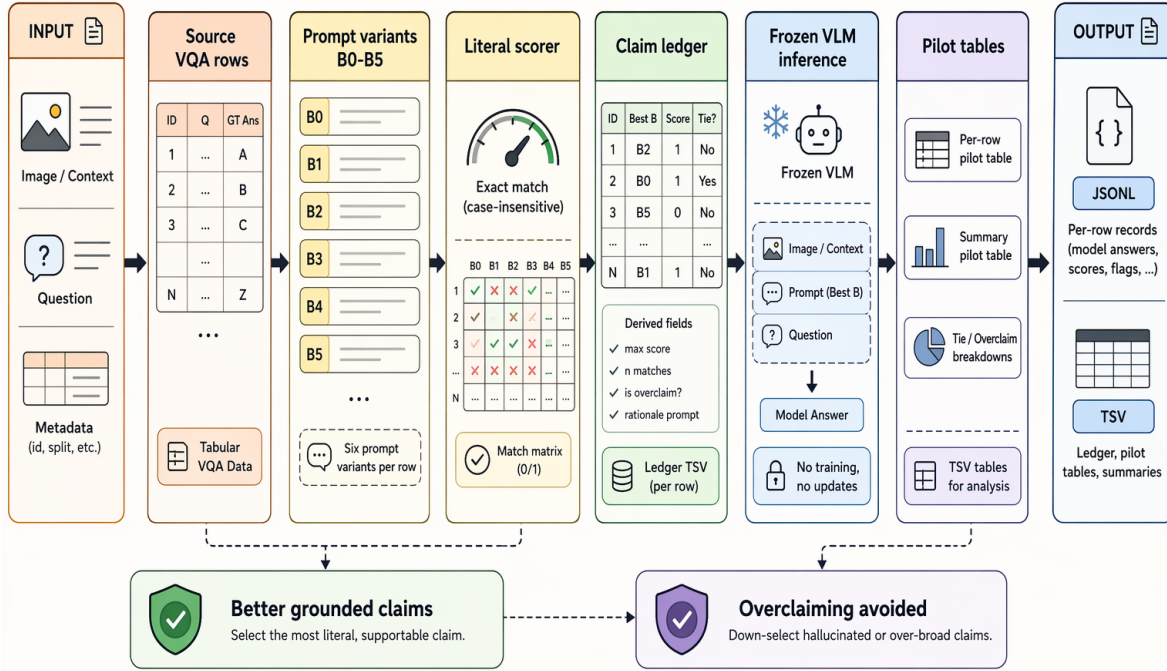


Figure 2: Train-free verification pipeline. Source rows are expanded into B0–B5 prompt variants, run through one frozen model, scored by family-specific deterministic metrics, and mapped into a claim ledger that separates supported observations from contradicted, missing, or unavailable claims.

from claiming that all retained rows require image evidence under the scorer.

B6 is a scoped comparator/control rather than a seventh prompt recipe in the main matrix. It replays the same 50 retained rows per family through the true image-input route while recording generated-token log-probability summaries from the backend. The emitted answer is still scored by the same family metric as B0–B5: exact literal-answer accuracy for TextVQA and VQA v2, and mask-hit accuracy for Ref-Spatial. The table label “scoped” means that B6 is available for the retained 150-row comparator slice and for descriptive paired contrasts with B0, B1, and B5, not that it supplies a trained reranker, a hidden oracle, or an isolated mechanism intervention.

The claim ledger maps measurements to paper claims. A claim is marked weak when the evidence supports a row-scoped observation but not a general result. It is contradicted when the executed rows disagree with the proposed statement. It is marked as missing when the intended measurement is outside the current executable evidence contract. This prevents citations from filling empirical gaps: prior work motivates the controls (Luo et al. 2025; Deng et al. 2025; Vo et al. 2025), but it does not supply transformed no-conflict counterfactuals, GQA headline evidence, or a mechanism proof for the executed B6 comparator/control.

Results

Table 1 is the main audit matrix. It reports every retained family-by-method cell for Qwen2.5-VL-3B-Instruct. The table uses family-specific metrics and bolds the best score within each family. The result should be read as a controlled diagnostic matrix, not as a leaderboard.

The TextVQA slice is the most favorable case for B5 relative to the matched-format baseline. B1 scores 0.720000 and B5 scores 0.800000, a +0.080000 descriptive delta. B0 reaches 0.820000, so the result does not isolate evidence grounding as the only useful prompt change. The correct reading is that the evidence-grounded prompt recovers six B1 failures while losing two B1 successes under this exact scorer, with the paired exact test used only as a descriptive guardrail.

The TextVQA result also shows why the direct baseline remains in the matrix. If the paper only compared B1 and B5, the B5 gain would look like a clean improvement. But B0 remains higher than B5, and B2 is lower than both B0 and B5. The more careful interpretation is that the matched-format instruction hurts this TextVQA slice relative to direct answering, while the evidence-grounded variant recovers some of that loss. That is a useful diagnostic finding because it separates format sensitivity from a broader claim that evidence wording is uniquely effective.

The VQA v2 slice shows why the audit keeps B4 and

Table 1: Three-family Qwen2.5-VL-3B audit matrix: B5 improves over B1 on TextVQA by 0.080000, is net zero versus B1 on VQA v2, and is -0.040000 versus B1 on RefSpatial; B6 is an executed scoped comparator/control with scores 0.820000, 0.900000, and 0.180000.

Family / split	n	Backend	Budget	Metric	B0	B1	B2	B3	B4	B5
TextVQA / Open Images validation	50	Qwen2.5-VL-3B	greedy/64	exact acc.	0.820	0.720	0.760	0.640	0.100	0.800
VQA v2 / COCO validation	50	Qwen2.5-VL-3B	greedy/64	exact acc.	0.900	0.880	0.920	0.820	0.220	0.880
RefSpatial benchmark / Location	50	Qwen2.5-VL-3B	greedy/64	mask hit	0.240	0.260	0.140	0.160	0.000	0.220

Family / split	B6 budget	B6	B5-B1	p
TextVQA / Open Images validation	scoped	0.820	+0.080	0.289
VQA v2 / COCO validation	scoped	0.900	0.000	1.000
RefSpatial benchmark / Location	scoped	0.180	-0.040	0.625

B0-B5 are the core prompt-condition matrix; the companion panel reports the executed B6 comparator/control and the paired B1-B5 descriptive exact test that was previously isolated as a separate float.

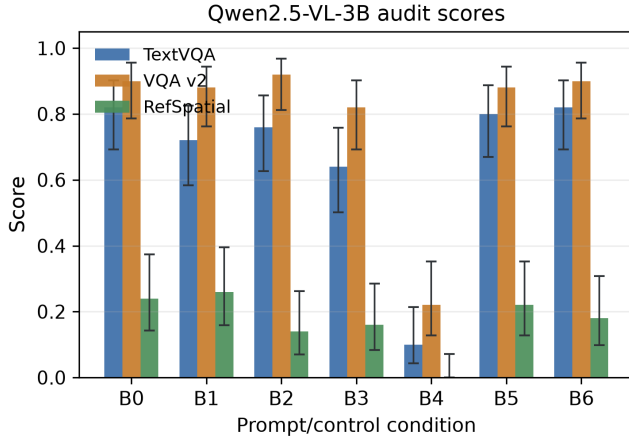


Figure 3: Family-specific B0-B6 scores for the Qwen2.5-VL-3B audit. Image-blind B4 remains lowest in each family, B6 is descriptive rather than mechanistic evidence, and TextVQA and VQA v2 still contain question-only successes.

paired counts visible. B1 and B5 both score 0.880000, with two B1-to-B5 recoveries and two B5 regressions. B4 scores 0.220000, so some VQA questions can be answered without the image. The result blocks a simple image-dependence statement for the whole VQA slice and blocks a broad B5 improvement claim.

The VQA v2 result is the clearest warning against overclaiming. The image-present rows are high across B0, B1, B2, and B5, and B2 reaches 0.920000 on the retained slice. However, eleven image-blind successes out of 50 show that a nontrivial part of this retained VQA slice is compatible with question-only answering. The paired B1-B5 counts further show that B5 is not monotonically safer: two rows are recovered and two rows are lost. Thus, the VQA evidence

supports a control-design lesson rather than a mitigation lesson. A future larger VQA pass should stratify by question type before treating the prompt effect as meaningful.

The RefSpatial slice is a hard case for this model and prompt family. B0 scores 0.240000, B1 scores 0.260000, B5 scores 0.220000, B6 scores 0.180000, B2 scores 0.140000, B3 scores 0.160000, and B4 scores 0.000000. This means the model often fails to produce a point that lands inside the target mask, even when the image is present. RefSpatial therefore supplies the third executable source family required for breadth, but it should not be presented as evidence that B5 improves spatial grounding or that B6 is a superior comparator.

RefSpatial is valuable despite the low score because it changes the interpretation of breadth. Without it, the paper would cover two exact-answer VQA-style families and would still be vulnerable to the objection that the matrix is only text or answer-string evaluation. With RefSpatial, the evaluation includes a spatial-reference task whose answer is a coordinate and whose scorer checks a mask. The result is negative for the proposed B5 prompt and for the scoped B6 control: neither improves over B1. That negative result is scientifically useful because it prevents the audit from being selected only around prompt successes.

The paired columns in Table 1 keep the prompt-effect claim calibrated. B5 produces nine B1-to-B5 recoveries and seven B5 regressions across the retained families. That balance supports row-level failure analysis, not a stable mitigation claim. The p-values are guardrails against overreading modest samples; they are descriptive checks rather than significance evidence.

The sample size reinforces the same point. At 50 rows per family, aggregate scores remain sensitive to a handful of changed answers. The TextVQA gain is a real observed row-level event, but the exact paired test is reported only as a de-

scriptive guardrail. The paper therefore reports exact paired counts and family-specific scores instead of converting the results into significance or robustness headlines. This is a deliberate presentation choice: the strongest contribution is the auditable matrix and its constraints, not a large numerical improvement.

Family-Level Interpretation

TextVQA provides the most favorable retained example for B5 relative to the matched-format baseline. The image-blind condition scores 0.100000, and the image-present conditions mostly score high. The retained questions are not trivially solved from wording alone, but they still do not prove careful image reading on every row. Direct answering scores 0.820000, evidence-grounded answering scores 0.800000, and matched-format answering is lower at 0.720000. The evidence points to a formatting-sensitive visible-text slice: a positive diagnostic for B5 relative to B1, plus a warning that the direct baseline already captures much of the same performance.

VQA v2 contributes a different kind of evidence. Its high image-present scores show that Qwen2.5-VL-3B handles many retained ordinary VQA questions, but the B4 score shows that some of those questions are also answerable from priors or question regularities. This is not surprising for VQA v2: many questions mention object categories, activities, and common scene relations that are statistically constrained by language and dataset composition. The important point is that the audit measures this instead of assuming it away. When B4 succeeds, an image-present success should be interpreted cautiously unless the row is inspected or stratified further.

The VQA v2 paired result also clarifies why aggregate equality is not the same as identical behavior. B1 and B5 both score 0.880000, but the paired table records two recoveries and two regressions. If only aggregate scores were reported, the prompt effect would appear to be zero and uninteresting. The paired counts show a more nuanced story: B5 changes at least four row outcomes, but the changes cancel in the aggregate (Deng et al. 2025; Vo et al. 2025). For a mitigation paper, that cancellation is a problem. For a diagnostic audit, it is useful evidence that the prompt is not merely a no-op.

RefSpatial is the strongest negative signal. The model almost never returns a point that falls inside the target mask, and B5 does not help. Because B4 is zero, the failure is not that the task is solved from the text alone. Instead, the evidence suggests that the current prompting interface is poorly matched to precise point localization, that the model may not reliably emit usable coordinates, or that the retained RefSpatial rows are too strict for a compact prompt (Li et al. 2025c; Liu et al. 2025; Li et al. 2025a). The audit cannot determine which explanation is correct. It can, however, keep the spatial failure visible so that the paper does not present only the VQA-style families.

Together, the three families motivate the paper’s methodological stance. A prompt intervention that looks plausible in one family can be neutral or ineffective in another; a high score can coexist with image-blind success; and a low score

can reflect scorer-interface mismatch as much as visual misunderstanding. The paper therefore avoids aggregate claims, robust-mitigation language, and model-internal mechanism claims.

Analysis and Failure Cases

The first lesson is that image-blind controls change the interpretation. RefSpatial has B4 at zero, TextVQA has five B4 successes out of 50, and VQA v2 has eleven B4 successes out of 50. The question distribution and common visual priors can therefore carry some rows, especially in VQA v2. A paper that omitted B4 would mistake all image-present success for visual evidence use.

This is especially important for prior-override claims. Prior use is not always an error: if a common object or scene relation is statistically reliable, a question-only answer can be correct for the wrong reason or for a reason the scorer cannot distinguish. The B4 condition is a coarse test, but it is still informative. It says whether the prompt alone contains enough information for the model to hit the target under the same normalization rule. The VQA v2 and TextVQA B4 results therefore weaken any claim that all retained rows are visually decisive, while the RefSpatial B4 zero makes that family a better candidate for image-dependence discussion under the strict point scorer.

The second lesson is that B5 has mixed row-level behavior. In TextVQA, the +0.080000 net gain over B1 is consistent with a prompt helping visible-text recovery, but it still includes two regressions. In VQA v2, B5 recovers two B1 failures and loses two B1 successes. In RefSpatial, B5 recovers one B1 failure and loses three B1 successes. The correct claim is therefore about diagnostic bookkeeping: evidence-grounded prompting changes some answers, but the current evidence does not show a stable improvement.

The row-level taxonomy also separates recovery from stability. A recovery is counted only when B1 is wrong and B5 is correct under the family metric; a regression is counted when B1 is correct and B5 is wrong. TextVQA has six recoveries and two regressions, VQA v2 has two recoveries and two regressions, and RefSpatial has one recovery and three regressions. These events are sufficient to show why aggregate equality can be misleading: VQA v2 ties in aggregate despite changed rows, while RefSpatial moves in the wrong direction.

The third lesson is that metric heterogeneity matters. Exact-answer VQA scoring and RefSpatial mask-hit scoring measure different skills. Pooling them would hide the fact that Qwen2.5-VL-3B is strong on the retained VQA-style slices and weak on the point-localization slice. The audit therefore keeps family-specific tables, family-specific metrics, and row-level evidence rather than reporting one average.

Metric heterogeneity also affects failure interpretation. A wrong TextVQA answer, VQA answer, or RefSpatial point can reflect different mixtures of recognition, normalization, prior use, parsing, target identification, or coarse localization (Luo et al. 2025; Deng et al. 2025; Li et al. 2025c). The present audit does not resolve those subtypes;

it records enough structure to prevent unlike failures from being treated as one phenomenon.

The claim discipline follows directly from these measurements. The paper can claim that a three-family diagnostic matrix was executed for a named frozen VLM over 50 tasks per retained family, plus a scoped B6 comparator/control over the same 150 retained rows. It cannot claim that B5 beats direct prompting, because B5 does not dominate B0 across families. It can report that B5 beats the matched-format baseline on TextVQA by 0.080000, but it must also report net-zero B1–B5 behavior on VQA v2 and negative B1–B5 behavior on RefSpatial. It can discuss channel differences between visible text, object-question, and spatial-reference rows, but it cannot claim a complete prior-channel taxonomy. The no-conflict evidence is a source-clean proxy rather than a transformed congruent control set, and B6 is descriptive rather than mechanism-isolating evidence.

Protocol Caveats

The first caveat is that the B0–B5 labels are experimental conditions, not method names in the usual algorithmic sense. B5 asks the model to prioritize visible evidence, but it adds no verifier, cropper, detector, learned reranker, or causal intervention. If B5 improves a row, the paper can say that the prompt condition changed the answer; it cannot say that a separate verification module found the relevant pixels.

The second caveat is that exact answer scoring is intentionally unforgiving. This is appropriate for visible text and short VQA answers when source annotations contain the target strings, but it can undercredit semantically equivalent responses. The choice is still defensible for this audit because the main question is comparative: the same scorer is applied across B0–B5 within each family. A semantic judge would introduce its own model identity and calibration problem. The present version therefore uses deterministic scoring and states the resulting limitation explicitly.

The third caveat is that RefSpatial’s mask-hit metric is not a proxy for all spatial reasoning. It checks whether one generated point lies inside one target mask. A model could understand the referred object but output a point near the boundary, output a box instead of a point, or describe the target in words. Those outputs still fail this scorer. The negative RefSpatial result should therefore be read as a failure under a strict point-output protocol, not as a complete statement about the model’s spatial knowledge.

The fourth caveat is row selection. The retained rows are those that could be replayed under a common evidence contract. That contract is necessary for honest evaluation, but it may bias the slice toward examples with available images, parseable questions, and scorer-compatible targets. The paper treats this as a scoped diagnostic audit. It would be incorrect to use the current rows to rank model families, to estimate benchmark-level accuracy, or to claim that B5 generalizes to prior-conflict perception broadly.

The fifth caveat concerns statistical testing. The paired exact tests use the row as the comparison unit, but with 50 rows per family they are descriptive guardrails rather than strong inferential evidence. The paper therefore uses row

counts and family-specific directionality rather than significance language.

Evidence Boundaries

The evidence boundary has two important components: the evaluated backend must be identifiable, and the retained source families must be executable under the same image-backed replay protocol. The present audit addresses both. Qwen2.5-VL-3B-Instruct is a reader-facing model identity, and RefSpatial supplies a third family with a different answer form and metric. Those choices make the audit interpretable as an empirical study of one frozen VLM.

They do not make the audit complete. The retained slices are best understood as a scoped diagnostic sample at the current executable source maximum. The rows are source-derived and replayed uniformly, but they are not enough to estimate a stable population effect for TextVQA, VQA v2, or RefSpatial. They also do not isolate a mechanism inside the model. A B5 recovery may reflect better visual checking, more cautious answer formatting, or a prompt-specific prior shift. A B5 regression may reflect overcorrection or a formatting error. The matrix records the event; it does not identify the internal cause.

The GQA exclusion is another boundary. The intended role of GQA/Visual Genome was scene-graph reasoning and relational prior conflict. The available rows for this pass did not meet the same image-backed replay standard, so the paper replaces that branch with RefSpatial rather than treating unavailable GQA rows as if they had been measured. This is a conservative choice. It gives the paper a third executable family, but it narrows the claim from scene-graph reasoning to spatial-reference localization.

Finally, the no-conflict guardrail is only a proxy. A prior-override method should ideally help when the prior is wrong and avoid damage when the prior agrees with the image. The current run includes source-clean rows summarized as a no-conflict proxy, but it does not include transformed congruent counterfactual controls. For this reason, the paper avoids language such as safe, robust, or generally faithful. The result is a measured audit of prompt behavior, not a deployment recommendation.

Comparator and Guardrail Scope

The executed B6 condition is useful because it closes a previous evidence gap, but it does not change the paper’s claim level. B6 is a scoped log-probability comparator/control run over the same 150 retained image-backed rows as the main three-family slice. It uses true image input and generated-token log-probability evidence, so it is now an executed branch. Its scores are 0.820000 on TextVQA, 0.900000 on VQA v2, and 0.180000 on RefSpatial. This pattern is descriptive: B6 matches B0 on TextVQA, matches B0 on VQA v2, and falls below B0, B1, and B5 on RefSpatial.

The paired B6 contrasts reinforce the same boundary. Against B5, B6 changes TextVQA by +0.020000, VQA v2 by +0.020000, and RefSpatial by -0.040000. Against B1, it changes TextVQA by +0.100000, VQA v2 by +0.020000, and RefSpatial by -0.080000. These values are useful checks

on whether a train-free comparator can be executed under the same replay contract. They are not evidence that B6 isolates a model-internal prior-suppression mechanism. The RefSpatial negative result is especially important: an audit that reported only the two VQA-style gains would overstate the comparator’s scope.

The source-clean guardrail branch answers a different question. It replays B1 and B5 on 300 source-clean rows, 50 tasks for each retained family and two prompt conditions. The measured B5-minus-B1 deltas are +0.080000 for TextVQA, 0.000000 for VQA v2, and -0.020000 for RefSpatial. These rows are useful because they check whether the evidence-grounded prompt damages ordinary source-clean examples in the same retained slices. They are still proxy evidence: they are not transformed congruent counterfactuals where the same image and question are edited so that prior and evidence agree. The paper therefore reports them as guardrails, not as a safety guarantee.

This distinction follows the related diagnostic literature: prior-conflict and modality-conflict studies motivate checking learned regularities (Luo et al. 2025; Deng et al. 2025; Vo et al. 2025), while language-prior and decode-time work motivates train-free controls (Wu et al. 2025; Zhao et al. 2025; Ren et al. 2026; Guo et al. 2026). None supplies the missing transformed controls for this run, so the current audit records executed comparator and guardrail rows without promoting them to unsupported conclusions.

Discussion

The audit is useful because it converts a tempting mitigation story into a measured evidence story. A result-first narrative would highlight the TextVQA B5 gain and hide the VQA and RefSpatial null effects. The ledger does the opposite. It preserves the TextVQA recovery, places it next to the VQA recovery/regression pair, and adds the RefSpatial failure mode. That presentation is less sweeping but more faithful to the evidence.

The named backend identity changes the reproducibility status of the study. The evaluated system is Qwen2.5-VL-3B-Instruct under frozen inference. A reader can therefore interpret the matrix as evidence about one named VLM and reproduce the protocol with the same public model family. This does not make the result broad; it makes the empirical limits visible rather than entangled with model identity (Wu et al. 2025; Luo et al. 2025).

The replacement third family also changes the evidence boundary. RefSpatial gives the paper an evaluable image-backed family with masks and a deterministic spatial metric. It does not replace GQA as scene-graph reasoning evidence, and the paper should not claim that it does. GQA remains excluded until the required source images can be evaluated under the same replay protocol. The honest statement is that the new run has three retained families, including one spatial-reference replacement, while the originally planned GQA branch remains unavailable.

The broader lesson is methodological. Train-free VLM mitigation papers are easy to write as prompt recipes, but the empirical burden is heavier than a before-and-after table. The evidence has to show which rows require the im-

age, which rows are format-sensitive, whether the proposed prompt recovers failures without creating regressions, and whether the task family changes the conclusion. This paper does not satisfy that full burden at scale, but it demonstrates the audit structure needed to ask those questions without hiding negative or missing results.

The most informative negative result is RefSpatial. If the evidence-grounded prompt were a general visual-grounding repair, it should have improved a task where the answer must be a location in the image. It did not. That failure suggests that the prompt may be better suited to answer selection in short VQA rows than to localization, or that Qwen2.5-VL-3B needs a more explicit coordinate or region interface for spatial tasks. The current evidence cannot choose between these explanations, so the paper states them as hypotheses for follow-up rather than as conclusions.

The most informative positive result is modest. TextVQA shows six B1–B5 recoveries and two B5 regressions. Because TextVQA B4 is far lower than the image-present conditions, those recoveries occur in a family where question-only answering solves few retained rows. This supports a narrow statement that evidence-grounded prompting can change visible-text answers in the desired direction under the retained protocol. It does not support a statement about reliable OCR improvement, because B0 remains higher than B5 and the paired exact test is only a descriptive guardrail.

The next evidence round should focus on stratification and stronger controls rather than prompt ornamentation. It should stratify TextVQA by visible text type, stratify VQA v2 by question type, and stratify RefSpatial by target size and relation type (Li et al. 2025c; Liu et al. 2025). It should add transformed no-conflict rows where image evidence and likely priors agree, because a mitigation that helps conflict rows but damages clean rows is not stable. It should also add a comparator interface that exposes a usable train-free contrastive signal if comparator claims are reintroduced (Zhao et al. 2025; Guo et al. 2026).

Conclusion

This audit evaluates Qwen2.5-VL-3B-Instruct on a train-free prior-override diagnostic with three executable image-backed families, six B0–B5 prompt conditions, and an executed scoped B6 comparator/control. The result provides a named-backend, three-family evidence matrix, but it does not support a broad mitigation claim. B5 improves the matched-format baseline by 0.080000 on TextVQA, is net zero on VQA v2, and is -0.040000 on RefSpatial.

The paper-facing contribution is a calibrated diagnostic protocol, a three-family evidence matrix, and a claim ledger that keeps unsupported prior-suppression claims out of the manuscript. The main positive result is not that B5 solves prior override. It is that the audit makes such a statement visibly unsupported, while preserving the narrower row-level observations that are actually measured. That is the standard a train-free prompt study should meet before it is scaled or reframed as a mitigation method.

Limitations

The first limitation is scale and selection. The run has 50 tasks per retained family, which is enough to inspect the protocol and expanded three-family evidence matrix, but still not enough for a robust benchmark claim. Confidence intervals remain wide enough that row composition matters.

The second limitation is metric scope. TextVQA and VQA v2 use exact literal-answer accuracy, while RefSpatial uses mask-hit scoring for generated points. These metrics are deterministic and source-grounded, but they should not be pooled into one aggregate. A future study should add semantic adjudication and region-level checks before making fine-grained grounding claims.

The third limitation is control scope. GQA/Visual Genome remains excluded because the required source images were not available in an evaluable form for this audit. The no-conflict guardrail is source-clean proxy evidence, not transformed congruent counterfactual evidence. B6 is executed as a scoped log-probability comparator/control with 150 rows, but it is not an isolated mechanism test and is weaker than B5 on RefSpatial. These are evidence boundaries, not prose gaps.

The fourth limitation is model scope. The audit names Qwen2.5-VL-3B-Instruct because a public model identity is necessary for scientific interpretation, but a single model cannot establish a model-family conclusion. Larger open-weight VLMs, proprietary VLMs, and models with different visual encoders may respond differently to the same B0–B5 prompts. Cross-model evidence should be added before claiming that the protocol characterizes VLMs generally.

The fifth limitation is behavioral observability. The audit records outputs and scores, not attention maps, hidden states, or causal interventions. A B5 recovery could reflect better use of image evidence, a different answer prior, a parsing-friendly output, or another prompt-induced behavior. The paper therefore uses mechanism language cautiously and treats B5 as an observed condition rather than as an explanation.

Ethical Considerations

This work evaluates a diagnostic protocol and does not introduce a deployed decision system. The main ethical risk is overclaiming. A prompt that improves one retained slice could be presented as a general mitigation for hallucination, bias, or prior dominance. The paper reduces that risk by reporting image-blind controls, paired recoveries and regressions, excluded-family status, proxy-guardrail scope, and the scoped status of the executed B6 comparator/control.

The reported scores should not be used to certify safety, fairness, accessibility, or reliability in high-stakes vision-language applications. The retained rows are modest in scale, the metrics are narrow, and the audit does not evaluate downstream user harms. Future work should include broader source coverage, task-specific harm analysis, and human evaluation where outputs affect people or safety-critical decisions.

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Reproducibility Checklist

The replay package records the Qwen2.5-VL-3B three-family run, raw model outputs, scored rows, source manifests, result tables, significance table, failure taxonomy, claim graph, evidence gaps, and experiment audit. The study is frozen inference only: no training, adapter update, checkpoint write, retrieval index, reinforcement learning, or verifier model is used.